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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 2

Team No.: 10

Project Title: Priority Scheduling

Team Members

Roll Number	Student Name	Signature
2520030054	P. Manisha	
2520030176	P. Renuka Chowdary	
2520030147	K. Hemalatha	

1. Abstract

Priority Scheduling is a CPU scheduling algorithm used in Operating Systems to manage the execution of processes based on their assigned priority. In this project, processes are given different priority values, and the CPU selects the process with the highest priority for execution. The project demonstrates both **preemptive and non-preemptive Priority Scheduling** and calculates important parameters such as **Waiting Time, Turnaround Time, Completion Time, and Average Waiting Time**. A simple simulation can be used to visualize the order in which processes are executed and understand how priority affects CPU allocation. The project helps in understanding how operating systems efficiently manage multiple processes and make scheduling decisions. It also highlights the advantages and limitations of Priority Scheduling, including the possibility of **starvation** for low-priority processes. Overall, this project provides a simple and practical understanding of CPU scheduling and its role in operating system process management.

2. Problem Statement

In a multitasking operating system, multiple processes may be ready to execute at the same time, making it necessary to decide which process should receive CPU time first. Priority Scheduling solves this problem by assigning a priority to each process and selecting processes based on their priority. The project aims to develop a simple simulation of Priority Scheduling that determines the execution order of processes and calculates Completion Time, Waiting Time, Turnaround Time, and Average Waiting Time. The system should demonstrate both preemptive and non-preemptive Priority Scheduling and show how different priority levels affect process execution.

3. Objectives

1. To understand the concept of **Priority Scheduling** in Operating Systems.

2. To assign different **priority levels** to processes.

3. To determine the **execution order** of processes based on their priority.

4. To calculate **Waiting Time, Turnaround Time, and Completion Time**.

5. To compare **Preemptive and Non-Preemptive Priority Scheduling**.

6. To analyze the performance of the scheduling algorithm using **Average Waiting Time**.

4. Proposed Methodology

*The project will be designed and implemented in **Linux** using C/C++. The system will accept process details such as Process ID, Arrival Time, Burst Time, and Priority. Based on the priority values, the scheduler will select processes for CPU execution. The project will implement **Preemptive and Non-Preemptive Priority Scheduling** and generate the execution order using a Gantt Chart. It will calculate Completion Time, Waiting Time, Turnaround Time, and Average Waiting Time. The results will be displayed clearly to compare the scheduling performance.*

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	**Purpose in the Project**
Process Scheduling	Determines the order of process execution
Priority Scheduling	Selects processes based on their priority
Process Management	Manages the execution of multiple processes
fork()	Can be used to create processes for simulation
wait()	Used to wait for child process completion
getpid()	Obtains the process ID during process simulation

6. Individual Contribution

Roll Number	**Student Name**	**Individual Responsibility**
2520030054	P. Manisha	Design and implement the Priority Scheduling algorithm
2520030176	P. Renuka Chowdary	Calculate scheduling parameters and generate Gantt Chart
2520030147	K. Hemalatha	Testing, output analysis, and project documentation

7. Tools / Platforms / Software Used

(List the programming languages, operating system, libraries, tools, and software planned for the project.)

Tool / Platform / Software	Purpose
Linux / Ubuntu	Operating system environment for developing and testing the project.
C / C++	Implementation of the Priority Scheduling algorithm
GCC Compiler	Compiling and executing the C/C++ program
VS Code / Text Editor	Writing and editing the source code

Terminal	Running the program and viewing the output

8. Expected Outcome

*The project is expected to successfully implement **Priority Scheduling** in a Linux environment. It will accept process details such as Process ID, Arrival Time, Burst Time, and Priority, and determine the order of process execution. The system will display the **Gantt Chart, Completion Time, Waiting Time, Turnaround Time, and Average Waiting Time**. The project will help demonstrate how priority-based CPU scheduling works and provide a clear understanding of process management in Operating Systems.*

9. GitHub Repository

GitHub Repository Link: https://github.com/manishaperugu49-glitch/2520030054_176_147_OSSP

Repository Created: Yes

☐ Yes ☐ No

Faculty Name and Signature

Faculty Name: Dr. Anusha

Signature:

Remarks:

Faculty Use

Project Approved: _____

☐ Yes ☐ No

Date: _____

Faculty Signature: _____